H M A Mohit *Chowdhury*

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Computer Science PhD candidate with specialization in deep learning architectures for genomics. Extensive experience in developing and scaling Self-Supervised Learning systems, Computer Vision models that work with large-scale and high-dimensional genomic time series data. Balances strong ML theory background with practical software engineering, using Python and cloud-native technologies to create scalable AI infrastructure for interpreting genomic sequences.

Education

Ph.D. in Computer Science and Engineering — **GPA: 4.00**
University of North Texas — *2025 - 2027 (Expected)*
University of Colorado Colorado Springs (Transferred to UNT in Fall 2025) — *2022 - 2025*
Dissertation Title: *Deep Learning algorithms for multiscale 3D genome organization analysis.*
Advisor: *Oluwatosin Oluwadare*

M.Sc. in Computer Science — **GPA: 3.867**
University of Colorado Colorado Springs — *2024*

B.Sc. in Computer Science and Engineering — **GPA: 3.71**
Ahsanullah University of Science and Technology — *2017*

Peer-reviewed Journal Articles (08) (Google Scholar)

- **Chowdhury, H. M. A. M.**, Oluwadare, O. *HiCInterpolate: 4D Spatiotemporal Interpolation of Hi-C Data for Genome Architecture Analysis.* bioRxiv (2026), https://doi.org/10.64898/2026.02.06.704438
- **Chowdhury, H. M. A. M.**, Oluwadare, O. *EmbedTAD Using Graph Embedding and Unsupervised Learning to Identify TADs from High-Resolution Hi-C Data.* Communications Biology 9, 7 (2026), https://doi.org/10.1038/s42003-025-09224-z
- **Chowdhury, H. M. A. M.**, Fuller, M., & Oluwadare, O. *Robin: an advanced tool for comparative loop caller result analysis leveraging large language models.* NAR Genomics and Bioinformatics 8(1), 2026, https://doi.org/10.1093/nargab/lqag009
- Pinchuk, D., **Chowdhury, H. M. A. M.**, Pandeya, A., & Oluwadare, O. *HiCForecast: dynamic network optical flow estimation algorithm for spatiotemporal Hi-C data forecasting.* Bioinformatics 41(2), 2025, https://doi.org/10.1093/bioinformatics/btaf030
- Rohit, M., **Chowdhury, H. M. A. M.**, & Oluwadare, O. *ScHiCAtt: Hi-C Super Resolution Using Attention Methods for Single Cell.* Computational and Structural Biotechnology Journal 27, 2024, https://doi.org/10.1016/j.csbj.2025.02.031
- **Chowdhury, H. M. A. M.**, Boult, T., & Oluwadare, O. *Comparative study on chromatin loop callers using Hi-C data reveals their effectiveness.* BMC Bioinformatics 25(1), 2024, https://doi.org/10.1186/s12859-024-05713-w
- Houchens, D., **Chowdhury, H. M. A. M.**, & Oluwadare, O. *coiTAD: Detection of Topologically Associating Domains Based on Clustering of Circular Influence Features from Hi-C Data.* Genes 15(10), 2024, https://doi.org/10.3390/genes15101293
- Akpokiro, V., **Chowdhury, H. M. A. M.**, Olowofila, S., Nusrat, R., & Oluwadare, O. *CNNSplice: Robust models for splice site prediction using convolutional neural networks.* Computational and Structural Biotechnology Journal 21, 2023, https://doi.org/10.1016/j.csbj.2023.05.031

Poster Presentation (05)

- HiCInterpolate: 4D Spatiotemporal Interpolation of Hi-C Data for Genome Architecture Analysis. (*ISMB 2026*)
- EmbedTAD Using Graph Embedding and Unsupervised Learning to Identify TADs from High-Resolution Hi-C Data. (*MCBIOS 2026 & Pikes Peak AI Summit 2025*)
- Comparative study on chromatin loop callers using Hi-C data reveals their effectiveness. (*RECOMB 2024 & MLRD Research Day 2024*)

Talk (02)

- EmbedTAD Using Graph Embedding and Unsupervised Learning to Identify TADs from High-Resolution Hi-C Data. (*MCBIOS 2026 & Pikes Peak AI Summit 2025*)

Professional Service (02)

- **Peer Reviewer (07)**: i. BMC Bioinformatics, ii. Scientific Reports, iii. Computational and Structural Biotechnology Journal (CSBJ), iv. Genome Biology, v. BioData Mining, vi. IEEE International Conference on Bioinformatics and Biomedicine (BIBM), and vii. Analytical Letters
- **Mentor (01)**: UR2PHD Program, University of North Texas (Fall 2025 – Spring 2026)

Skills

- **AI/ML**: Deep Learning, Time-Series, Spatiotemporal Modeling, Transformer Architecture, Diffusion Architecture, CNNs, Graph Representation, Clustering.
- **Programming Languages**: Python, Java, C#, C, JavaScript
- **ML & Data Framework**: PyTorch, RAPIDS (cuML, cuDF, etc.), scikit-learn
- **Software Framework**: Spring, .NET, Flask
- **Cloud & Deployment**: AWS (EC2, S3, etc), Docker, Terraform, Git
- **Databases**: MySQL, MSSQL, PostgreSQL
- **Operating Systems**: Linux, macOS, Windows

Experience (Total: 9 years; Research: 4 years + Industry: 5 years)

Graduate Research Assistant — **2022 - Present**
CompBio-AI Lab

- Developed advanced AI/ML and deep learning algorithms for genomic data, such as CNNs, graph embeddings, time series, and spatiotemporal models for Hi-C data analysis. Developed scalable data pipelines, algorithmic benchmarking, and deployed web-based tools with Python, Flask, and Docker.

Solutions Developer — **2022 – 2022**
Otto International

- Developed scalable AWS architecture with Terraform for ESB deployments, optimized Java/PostgreSQL-based services, and established Jenkins-based CI/CD pipelines while working with cross-functional teams to integrate enterprise applications.

Software Engineer — **2020 – 2022**
Dovetail Technology

- Designed and developed scalable applications using C#, .NET Web API, and MSSQL, deployed using IIS, developed microservices-based Windows services, and worked with other teams to enable the integration of AngularJS and improve workflow.

Software Engineer — **2019 – 2020**
IdeaScale Bangladesh

- Improvement of backend services with Java, Spring, MySQL, and Docker; backend stability with code refactoring and bug fixes; modernization of legacy SQL with JPA and Querydsl; and working cross-functionally to support the evolution of the scalable system.

Software Engineer — **2017 – 2019**
Netweaver Software Limited

- Led the development of two e-commerce platforms with Java, Spring Boot, MySQL, SAP UI5, and JavaScript, and a healthcare directory platform with Node.js, Vue.js, and PostgreSQL, focusing on scalable backend development and integration with the frontend.

Awards and Certificates

– MCBIOS 2026 Travel Award
– CENG Anticipated Award, 2025 - University of North Texas
– Graduate School Travel Award, 2024 - University of Colorado Colorado Springs
– Tuition Matching Grant Award, 2022-2024 - University of Colorado Colorado Springs
– Tableau Fundamentals
– Responsible Conduct of Research (RCR-mini)

Leadership and Activities

– Participant, Open House 2025 at UCCS
– President, Bangladeshi Student Association (BSA) 2024-2025 at UCCS
– Participant, Cool Science Festival 2022, 2023, 2024 at UCCS
– Session Chair, NSF REU Summer Research 2023 at UCCS
– Technical Assistant, NSF REU Summer Research 2023 at UCCS
– Conducted a basic Linux and Docker hands-on class in NSF REU Summer Research 2023 at UCCS
– Organized Cumilla Victoria Govt. College Reunion 2016